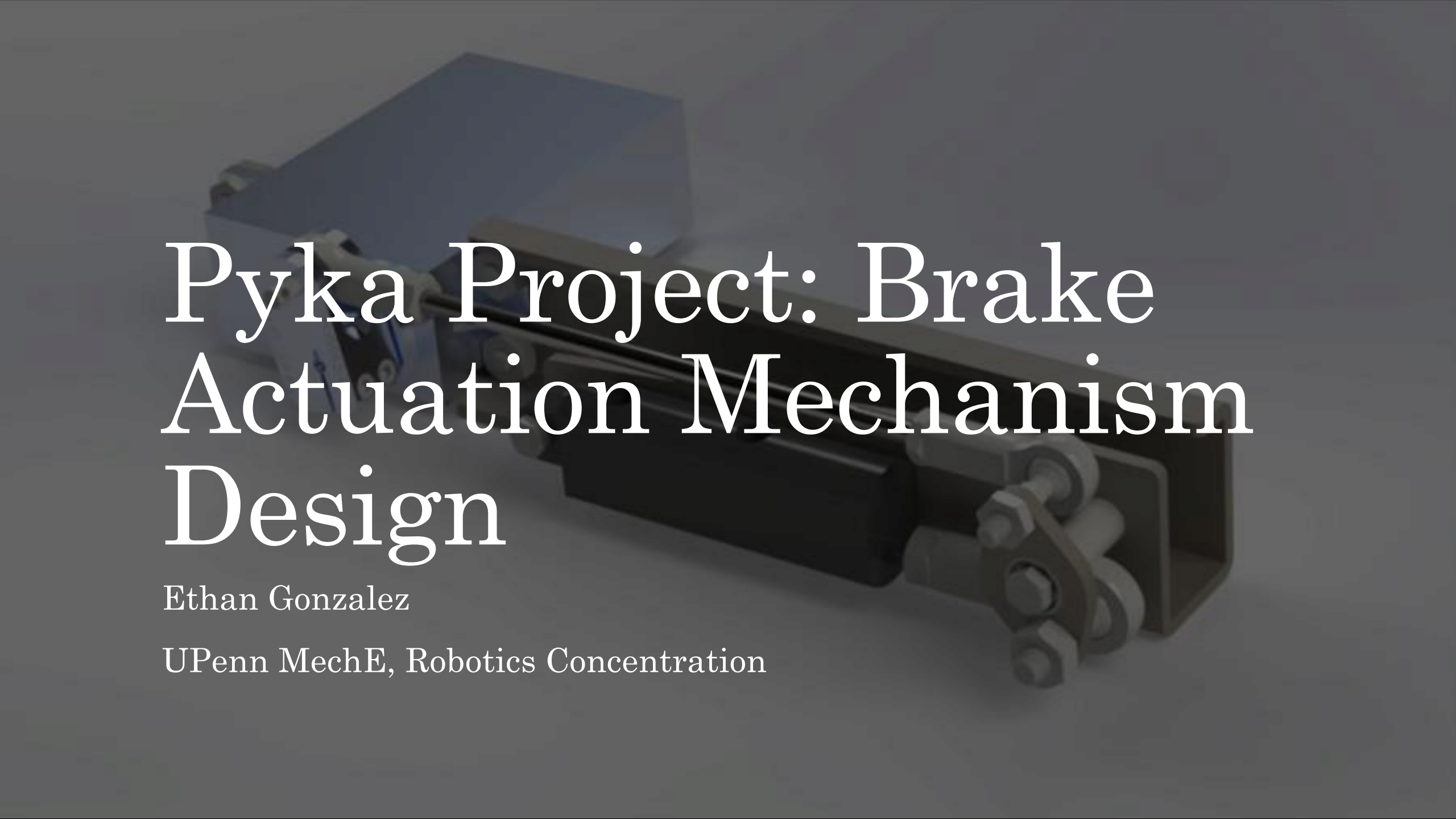


# Pyka Project: Brake Actuation Mechanism Design

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# About Me

I'm a mechanical engineer from Alameda who loves getting hands on, solving complex problems, and continuously learning, especially in the fields of robotics and propulsion. I'm pursuing a BSE in Mechanical Engineering and Applied Mechanics at the University of Pennsylvania, with a concentration in Dynamics, Controls, and Robotics.

With experience as Design Project Lead at Project Wavefront, Founder of Educate and Elevate, and Manufacturing Engineering Intern at Astra Space, I've developed a strong foundation in both technical and leadership skills. I know how to define ambitious goals, build a structured path to achieve them, and actively look for places where I can contribute and grow.

What motivates me most is the chance to collaborate with talented and curious minds from a wide range of fields. Outside of engineering, I'm an avid skiier, rock climber, and mountain biker. Nice to meet you!

# Brake Actuation Force Analysis

- Current horn radius (0.750”) cannot deliver 200 lbf at the master cylinder
- At stall torque, predicted output is ~170 lbf → deficit ~30 lbf
- Options: reduce effective horn radius to ≤ 0.63”, or add a lever with mechanical advantage ≈ 1.176 while preserving 0.2” stroke
- With MA = 1.176, required servo-side stroke is ~0.235”, which corresponds to ~0.313 rad (~17.9°) of servo rotation at a 0.750” horn radius
- Design options: lever, rocker arm, bell crank selected based on packaging envelope

## Brake Actuation Analysis

### Servo (Direct Drive)

**Given**

$$\tau_{stall} = 2039.22 \text{ oz-in} = 127.45 \text{ lbf-in}$$
$$r_h = 0.75 \text{ in}$$
$$F_{req} = 200 \text{ lbf}$$

**Max Output Force**

$$F_{out} = \frac{\tau}{r_h} = \frac{127.45}{0.75} = 169.9 \text{ lbf}$$

$$169.9 < 200$$

**Force Deficit**

$$200 - 169.9 = 30.1 \text{ lbf}$$

**Required Radius**

$$r_{req} = \frac{\tau}{F_{req}} = \frac{127.45}{200} = 0.637 \text{ in}$$

### Lever (Mechanical Advantage)

**Given**

$$F_{in} = 170 \text{ lbf}$$
$$F_{out} = 200 \text{ lbf}$$
$$x_{out} = 0.20 \text{ in}$$
$$r_h = 0.75 \text{ in}$$

**Required Ratio**

$$\frac{a}{b} = \frac{F_{out}}{F_{in}} = 1.176$$

**Stroke Relation**

$$x_{in} = x_{out}\frac{a}{b} = 0.235 \text{ in}$$

**Servo Rotation**

$$\Delta\theta = \frac{x_{in}}{r_h} = 0.313 \text{ rad} = 17.9^\circ$$

**Work Check**

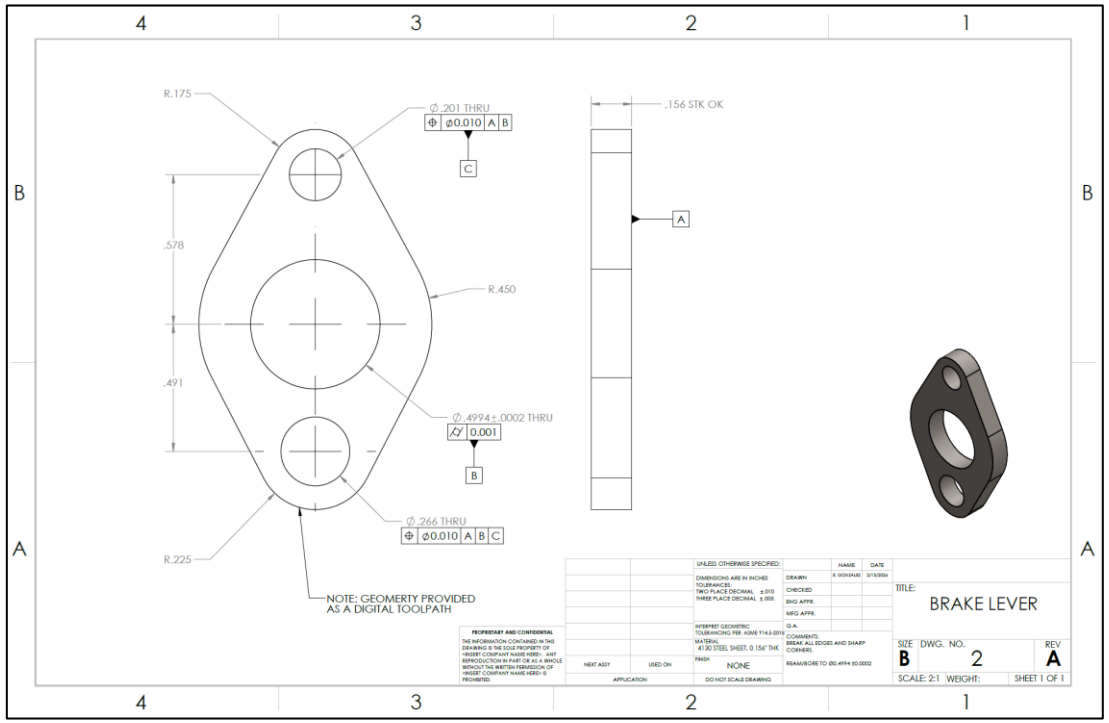
$$F_{in}x_{in} \approx F_{out}x_{out}$$

# Bill of Materials

- Materials
  - 5/32” 4130 steel sheet
  - 1/8″ 4130 or 1018 steel sheet
- Linkage
  - Steel Ball Bearing (Open, Trade Number R3, for 3/16" Shaft Diameter)
  - Ball Joint Linkage (1/4"-28 Internal Thread, Right-Hand Shank, Right-Hand Ball Stud)
  - Ball Joint Rod End, 10-32 Thread (Right-Hand)
  - Connecting Rod (Zinc-Plated Carbon Steel, 6" Overall Length, 10-32 Thread)
  - Steel Ball Bearing (Open, Trade Number R3, for 3/16" Shaft Diameter)
  - High-Strength 932 Bearing Bronze Sleeve Bearing (for 3/16" Shaft Diameter and 1/4" Housing ID, 1/4" Long)
  - High-Strength 932 Bearing Bronze Sleeve Bearing (for 1/4" Shaft Diameter and 3/8" Housing ID, 7/8" Long)
  - 1004-1045 Carbon Steel Clevis Pin (3/16" Diameter, 11/16" Usable Length)
- Fasteners
  - Zinc Yellow-Chromate Plated Hex Head Screw (Grade 8 Steel, 10-32 Thread Size, 1" Long)
  - Zinc Yellow-Chromate Plated Hex Head Screw (Grade 8 Steel, 10-32 Thread Size, 1-1/2" Long)
  - Zinc-Plated Alloy Steel Socket Head Screw (M3 x 0.5 mm Thread, 20 mm Long)
  - 18-8 Stainless Steel Hex Head Screws (12-24 Thread Size, 1/2" Long)
  - Corrosion-Resistant 18-8 Stainless Steel Hex Nuts (12-24 Thread Size)
  - Corrosion-Resistant 18-8 Stainless Steel Hex Nuts (10-32 Thread Size)
  - Corrosion-Resistant 18-8 Stainless Steel Hex Nut (1/4"-28 Thread Size)

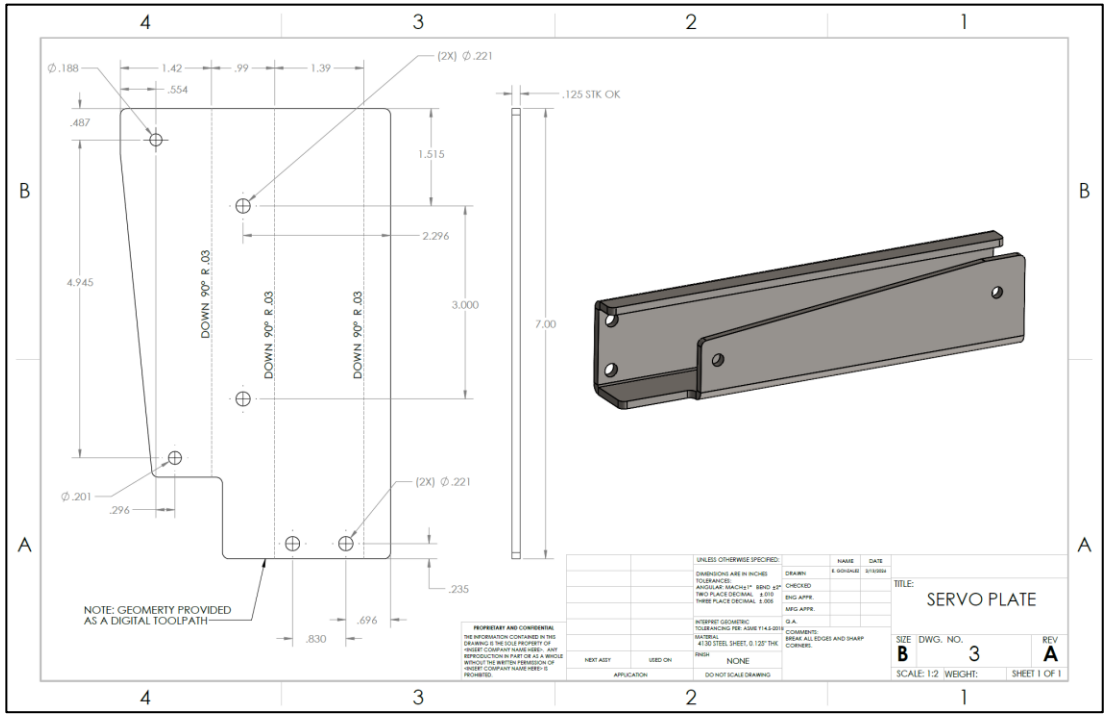
# Lever Design

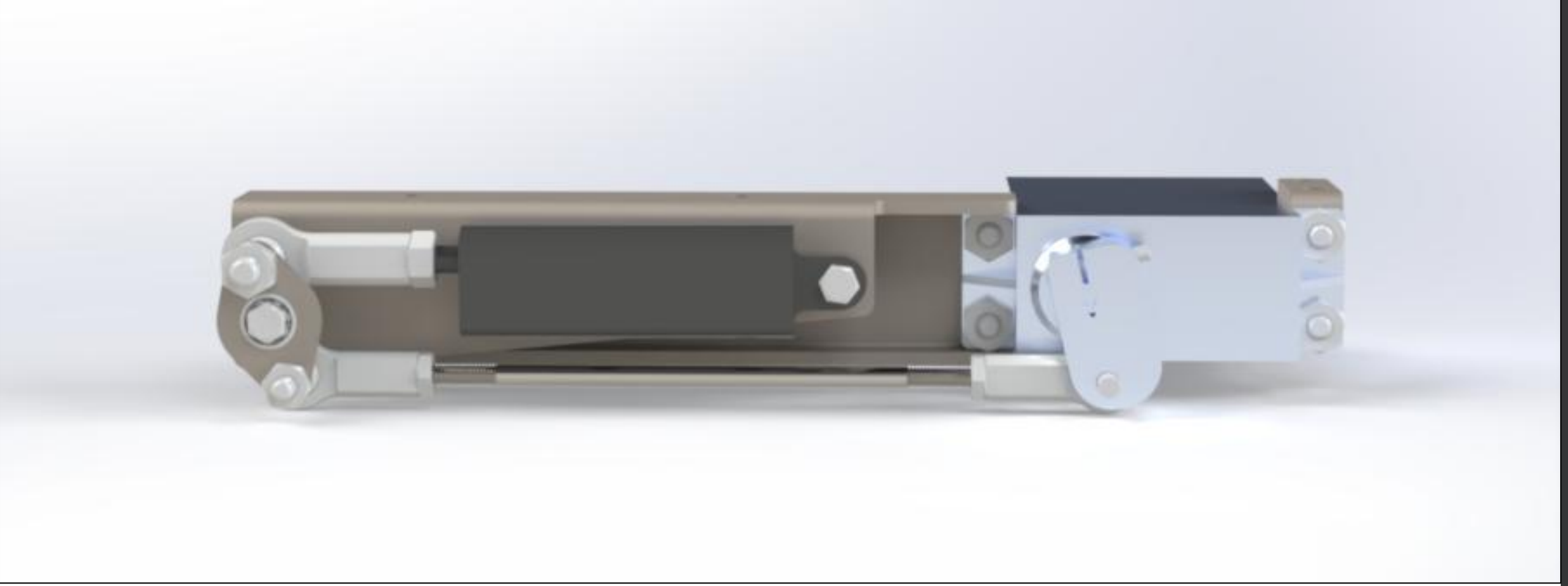
- 5/32” 4130 steel sheet selected for high strength and stiffness with laser/waterjet manufacturability
- 1.0694″ center-to-center hole spacing
- Lever arms: 0.578” (long) and 0.491” (short)
- Mechanical advantage = 0.578 / 0.491 = 1.176
- 180° force redirection enables compact packaging
- Designed for ball joint interfaces to accommodate misalignment
- Pivot supported by bearing for low-friction rotation



# Servo Plate Design

- 1/8” 4130 or 1018 steel sheet, waterjet/laser cut and formed via press brake or box-and-pan brake
- Designed to mount to a flat, vertically oriented bulkhead per assignment specification
- Self-contained structural frame; does not rely on bulkhead for stiffness, alignment, or load transfer
- Maintains profile within servo envelope for compact packaging
- Geometry optimized for bending stiffness and torsional rigidity under 200+ lbf loading
- Sheet-metal construction minimizes weight while preserving structural integrity
- No CNC required; cost-effective, scalable, and manufacturable at prototype or production volume
- Allows independent bench testing of full actuation assembly

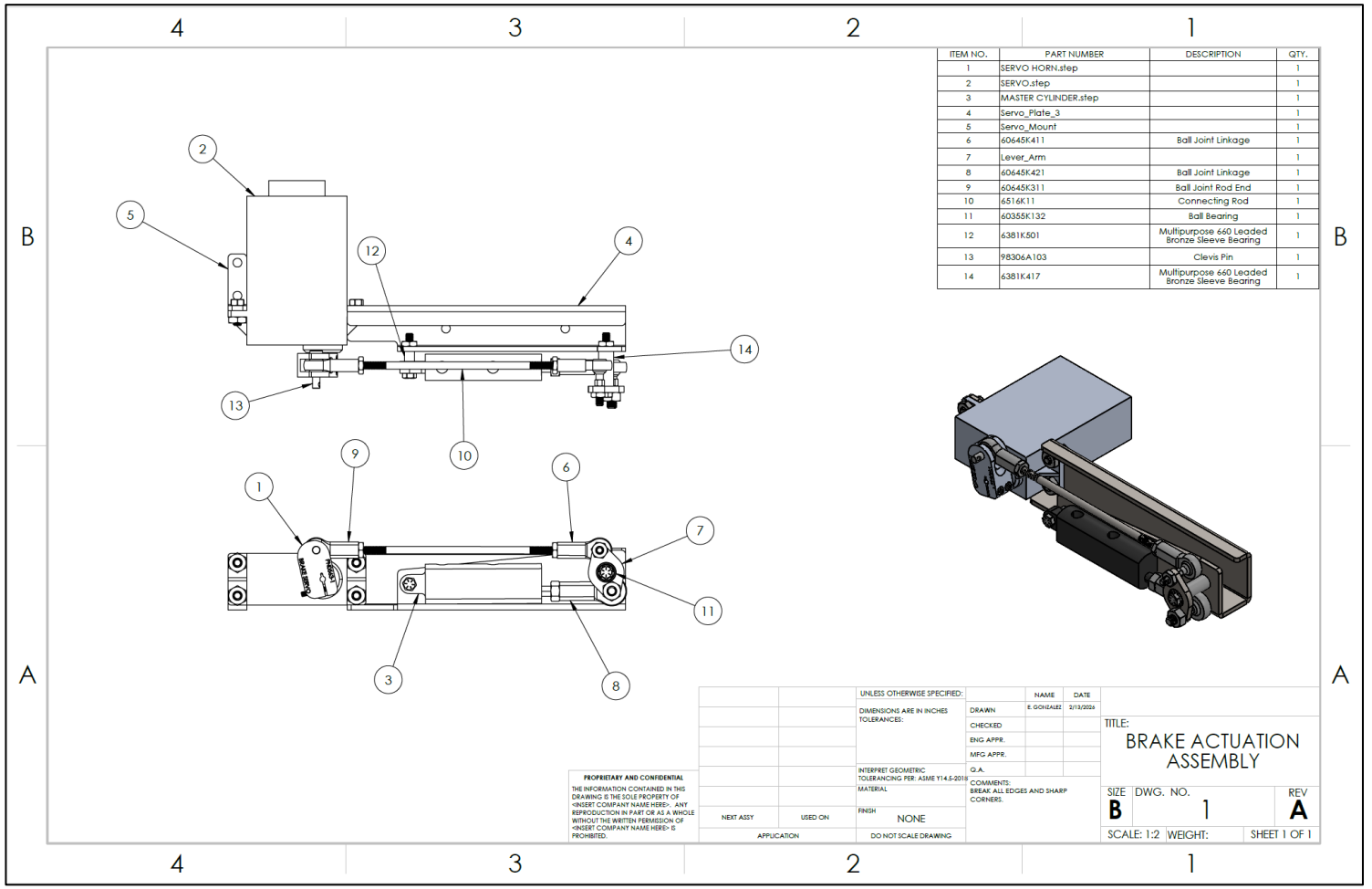




# Linkage

- All hardware sourced from McMaster to minimize cost and enable scalable production
- Linkage components include connecting rods, ball joint rod ends, ball bearings, clevis pins, and sleeve bearings
- Selected components have verifiable specification sheets
- Geometry maintains the smallest practical profile within supplied part constraints

# Assembly Drawing



| ITEM NO. | PART NUMBER | DESCRIPTION | QTY. |
|---|---|---|---|
| 1 | SERVO HORN.step | | 1 |
| 2 | SERVO.step | | 1 |
| 3 | MASTER CYLINDER.step | | 1 |
| 4 | Servo_Plate_3 | | 1 |
| 5 | Servo_Mount | | 1 |
| 6 | 60645K411 | Ball Joint Linkage | 1 |
| 7 | Lever_Arm | | 1 |
| 8 | 60645K421 | Ball Joint Linkage | 1 |
| 9 | 60645K311 | Ball Joint Rod End | 1 |
| 10 | 6516K11 | Connecting Rod | 1 |
| 11 | 60355K132 | Ball Bearing | 1 |
| 12 | 6381K501 | Multipurpose 660 Leaded Bronze Sleeve Bearing | 1 |
| 13 | 98306A103 | Clevis Pin | 1 |
| 14 | 6381K417 | Multipurpose 660 Leaded Bronze Sleeve Bearing | 1 |

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN INCHES
TOLERANCES:
INTERPRET GEOMETRIC TOLERANCING PER: ASME Y14.5-2018
MATERIAL
FINISH NONE
DO NOT SCALE DRAWING
NEXT ASSY | USED ON | APPLICATION

| | NAME | DATE |
|---|---|---|
| DRAWN | E. GONZALEZ | 2/12/2026 |
| CHECKED | | |
| ENG APPR. | | |
| MFG APPR. | | |
| Q.A. | | |

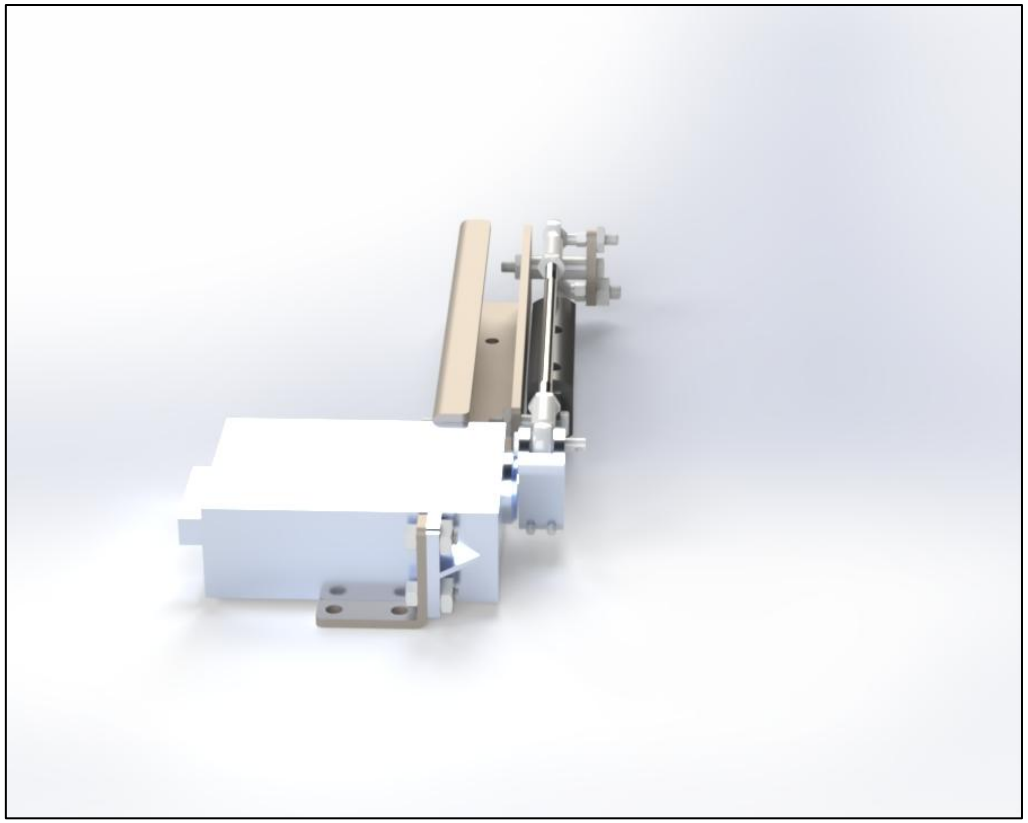
COMMENTS:
BREAK ALL EDGES AND SHARP CORNERS.

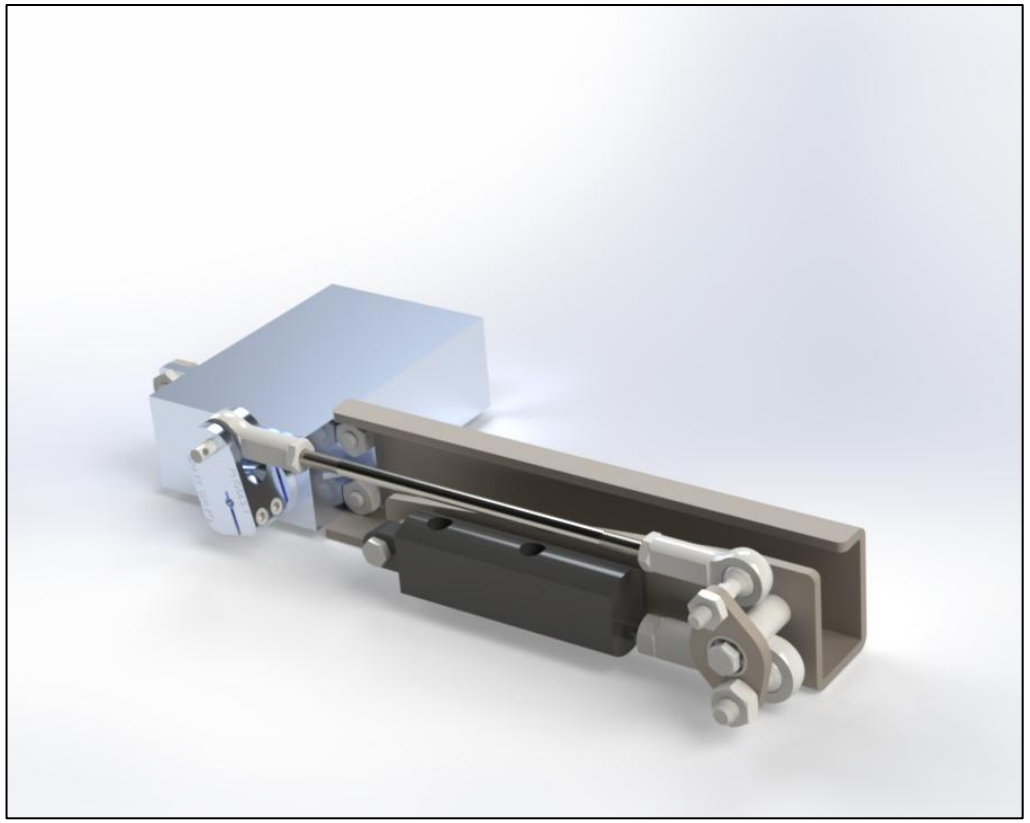
TITLE:
BRAKE ACTUATION ASSEMBLY

SIZE B | DWG. NO. 1 | REV A
SCALE: 1:2 | WEIGHT: | SHEET 1 OF 1

PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF <INSERT COMPANY NAME HERE>. ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT THE WRITTEN PERMISSION OF <INSERT COMPANY NAME HERE> IS PROHIBITED.

# Assembly Renders





# Connecting Rod Buckling Analysis

## Buckling Analysis: 3/16" × 6" Steel Rod (Fixed–Fixed)

**Given:**

$$d = 0.1875 \text{ in}, \quad L = 6 \text{ in}$$

$$P = 200 \text{ lbf}, \quad E = 29 \times 10^6 \text{ psi}$$

$$K = 0.5$$

**Section Properties:**

$$A = \frac{\pi d^2}{4} = 0.0276 \text{ in}^2$$

$$I = \frac{\pi d^4}{64} = 6.07 \times 10^{-5} \text{ in}^4$$

**Euler Buckling:**

$$P_{cr} = \frac{\pi^2 EI}{(KL)^2} = 1929 \text{ lbf}$$

**Buckling FOS:**

$$FS_{buckling} = \frac{P_{cr}}{P} = 9.65$$

**Compressive Stress:**

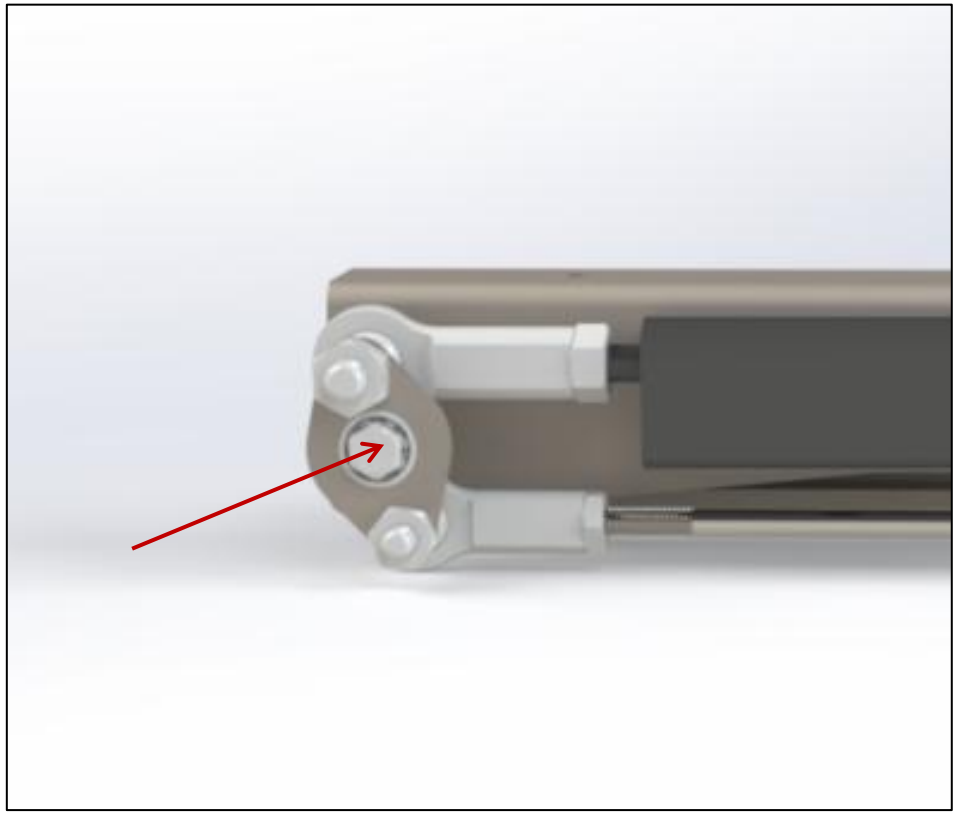
$$\sigma = \frac{P}{A} = 7240 \text{ psi}$$

**Yield Check (A36, 36 ksi):**

$$FS_{yield} = \frac{36,000}{7,240} = 4.97$$

Rod is in compression. Yield governs, but both yield and buckling are safe at 200 lbf.

# Pivot Bolt Shear Analysis



- **Other considerations:** Fatigue life assessment via S-N analysis under cyclic loading

### Shear Analysis: 10-32 Grade 8 Bolt (Single Shear)

**Given:**

$$V = 370 \text{ lbf}$$

$$d = 0.190 \text{ in}, \quad n = 32 \text{ TPI}$$

$$S_y \approx 130 \text{ ksi (Grade 8)}$$

**Tensile Stress Area (threaded section):**

$$A_t = \frac{\pi}{4}\left(d - \frac{0.9743}{n}\right)^2$$

$$A_t = \frac{\pi}{4}\left(0.190 - \frac{0.9743}{32}\right)^2 = 0.0200 \text{ in}^2$$

**Shear Stress (single shear):**

$$\tau = \frac{V}{A_t} = \frac{370}{0.0200} = 18{,}500 \text{ psi} = 18.5 \text{ ksi}$$

**Shear Yield Strength (von Mises criterion):**

$$\tau_y = \frac{S_y}{\sqrt{3}} = \frac{130}{\sqrt{3}} = 75 \text{ ksi}$$

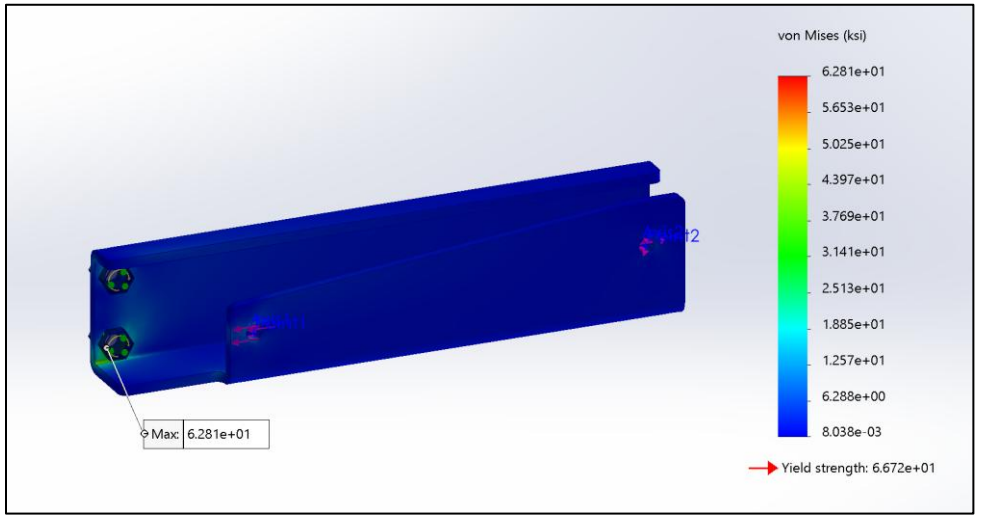
**Factor of Safety:**

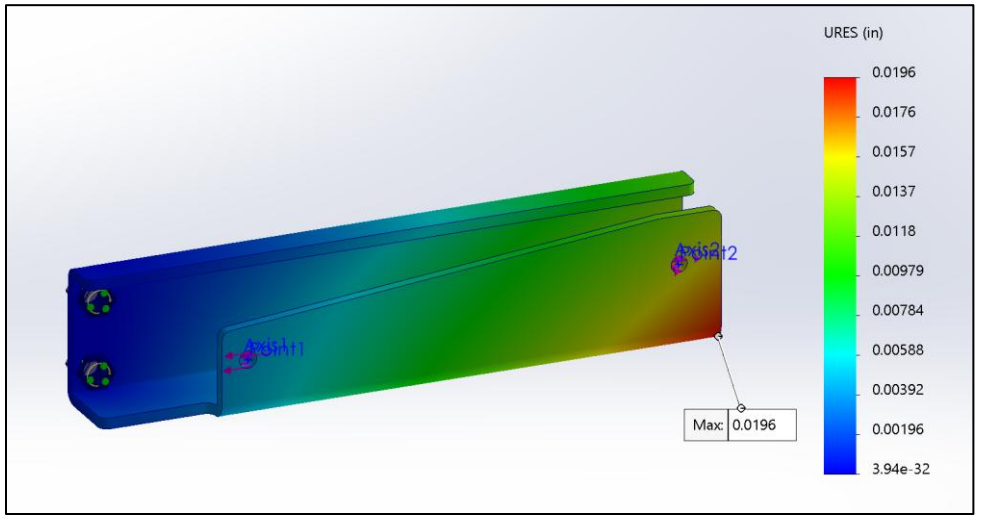
$$FS = \frac{\tau_y}{\tau} = \frac{75}{18.5} = 4.05$$

Bolt is safe in single shear at 370 lbf (FS ≈ 4.0)

# Servo Plate FEA

- Max displacement under loading conditions: 0.0196 in
- Yield stress (4130): 66.72 ksi
- Max stress: 62.81 ksi (200 lbf load includes FOS)
- Validates hole bearing stress and stress concentrations at hole edges
- 35 lbf-in bolt preload (per NASA/TM-2017-219475)
- 370 lbf bearing load @ pivot hole, 200 lbf bearing load @ cylinder hole



# Recommendations & Next Steps

- **Reduce Servo Horn Radius:** Eliminate added linkage and mechanical advantage to simplify the system and reduce part count
- **Optimize Servo Plate Weight:** Add lightening cutouts in low-stress regions of the servo plate with minimal added complexity using waterjet/laser cutting
- **Expand Fatigue Analysis:** Perform cyclic loading (S-N) analysis on pivot, lever, and linkage components
- **Validate with Testing:** Correlate hand calcs and FEA with bench testing at 200 lbf over 0.2” stroke